

2.28M
Total Children

8
Counties with Zero
Providers

67
Severe Desert Counties

23K
Children Without Local Access

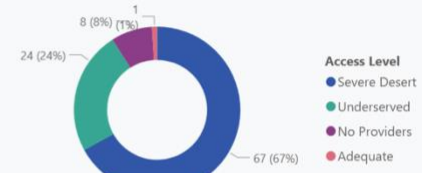
Region
☐ Coastal Plain
☐ Piedmont
☐ WNC

Pediatric Specialist Access by County

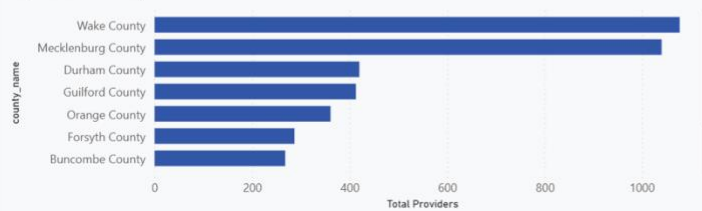
Access Level ● Adequate ● No Providers ● Severe Desert ● Underserved



Counties by Access Tier

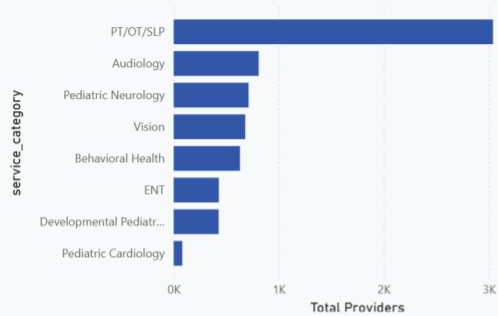


Top 10 Counties by Provider Count



County Name	Region	Child Population	Total Providers	Children Per Provider
Beaufort County	Coastal Plain	8748	7	1,249.71
Bertie County	Coastal Plain	3038		
Bladen County	Coastal Plain	6045	4	1,511.25
Brunswick County	Coastal Plain	20223	33	612.82
Camden	Coastal	2510	1	2,510.00
Total		2283888	6821	334.83

NC Pediatric Specialists by Type



Specialty Access

Region
☐ Coastal Plain
☐ Piedmont
☐ WNC

57
Behavioral Health
Deserts

85
Cardiology Deserts

50
Developmental Pediatrics Deserts

43
Pediatric Neurology Deserts

County Name	Audiologist	Behavioral Health Specialists	Developmental Pediatricians	Ear, Nose & Throat (ENT)	Pediatric Cardiologists	Pediatric Neurologists	PT / OT / SLP	Vision Care
Mecklenburg County	113	59	32	56	32	134	295	92
Durham County	42	23	12	23	10	56	99	68
Pitt County	37	11	14	6	9	20	80	13
Orange County	36	28	5	32	7	51	90	21
Wake County	119	44	157	56	7	64	347	81
Buncombe County	53	22	7	16	3	35	56	34
Catawba County	18	3	0	14	3	9	33	17
Cumberland County	34	9	18	17	3	30	67	23
Forsyth County	31	11	10	24	2	47	94	26
New Hanover County	22	10	9	13	2	30	66	38
Cabarrus County	12	7	9	7	1	30	34	14
Nash County	5	3	2	5	1	4	9	1
Total	810	316	429	423	83	702	2080	682

Western NC Pediatric Access

18 mountain counties. 554 pediatric specialists. Buncombe County (Asheville) holds 268 of them. 48% of regional supply concentrated in one urban hub while rural counties face severe access deserts. Hurricane Helene (Sept 2024) further disrupted access infrastructure across the region.

18

WNC Counties

554

Total Providers

48%

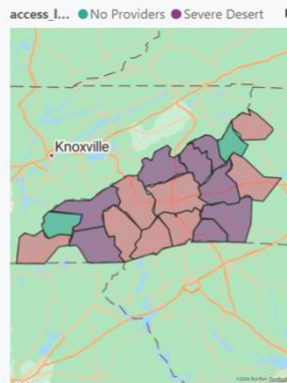
Buncombe Co. Share of WNC

2

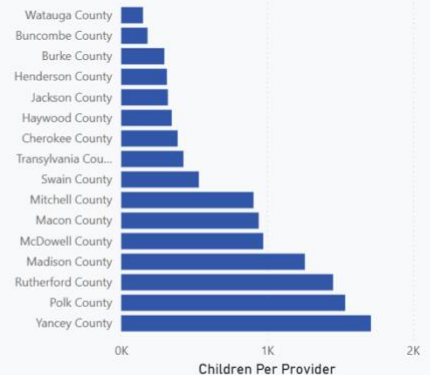
Counties with Zero Providers

County Name	Total Providers	Children	Children Per Provider	Impacted by Helene
Avery County		2455		Yes
Buncombe County	268	48538	181.11	Yes
Burke County	54	16016	296.59	Yes
Cherokee County	12	4646	387.17	No
Graham County		1637		No
Haywood County	32	11103	346.97	Yes
Henderson County	69	21666	314.00	Yes
Jackson County	22	7055	320.68	Yes
Macon County	7	6600	942.86	No
Madison County	3	3777	1,259.00	Yes
Total	554	169468	305.90	

Pediatric Specialist Access by County



Children Per Provider by WNC County



Methodology & Data Sources

This report integrates federal healthcare provider data with Census demographic data to map pediatric specialist access across all 100 NC counties. The analysis classifies counties into access tiers based on the ratio of children to specialists, identifying severe shortage areas where care delays are most likely.

Data Sources

- CMS National Provider Identifier (NPPES) Registry
 - 7,279 NC pediatric-relevant providers
 - Filtered to 13 healthcare provider taxonomy codes
 - <https://npiregistry.cms.hhs.gov>
- Census American Community Survey (ACS) 2022 5-Year Estimates
 - Child population (under 18) per county
 - Child poverty rates per county
 - <https://www.census.gov/programs-surveys/acs>
- Census TIGER/Line Shapefiles 2022
 - County boundary geometries
 - <https://www.census.gov/geographies/mapping-files>
- Census ZCTA-County Relationship File

Methodology

- Provider Supply
 - Queried NPPES API for NC providers across 13 pediatric-relevant taxonomy codes.
 - Mapped provider ZIP codes to county FIPS via Census crosswalk. Aggregated to county-specialty grain.
- Demand Calculation
 - Pulled child population (under 18) and poverty rates from Census ACS. Joined to provider data on county FIPS code.
- Access Classification
 - Calculated children-per-provider ratio per county. Classified counties into five tiers:
 - Well-Served: 50 or fewer children per provider
 - Adequate: 51 to 150
 - Underserved: 151 to 500
 - Severe Desert: 501 to 5,000
 - No Providers: zero specialists of any kind
- Regional Analysis
 - Defined Western NC pilot region as 18 mountain/Appalachian counties. Compared regional metrics against statewide and national benchmarks.

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GitHub: github.com/JohnApelU/nc-pediatric-access-pipeline